

RAJALAKSHMI ENGINEERING COLLEGE

Experiment 1: Setting Up the Python Environment and Libraries - Jupyter Notebook

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AIM

To verify the Jupyter Notebook environment by importing essential Python libraries, displaying an interactive widget, and creating a basic Matplotlib chart.

SOURCE

EXP1/Expt-01.ipynb

PYTHON CODE - CELL 1

```
import ipywidgets as widgets
from IPython.display import display

slider = widgets.IntSlider(
    value=50,
    min=0,
    max=100,
    step=1,
    description="Value:"
)

output = widgets.Output()

def on_value_change(change):
    output.clear_output()
    with output:
        print(f"Slider value is {change['new']}")

slider.observe(on_value_change, names="value")

display(slider, output)
```

EXECUTED OUTPUT

```
IntSlider(value=50, description='Value:')
Output()
```

PYTHON CODE - CELL 2

```
import matplotlib.pyplot as plt

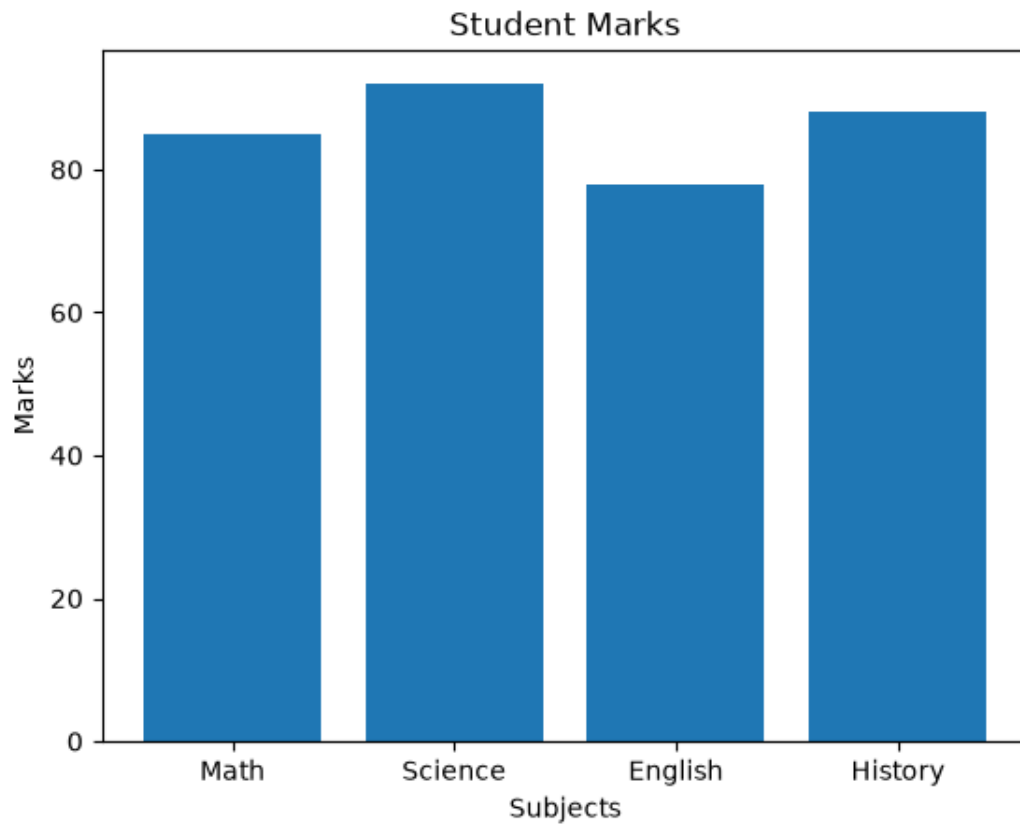
subjects = ["Math", "Science", "English", "History"]
marks = [85, 92, 78, 88]

plt.bar(subjects, marks)

plt.title("Student Marks")
plt.xlabel("Subjects")
plt.ylabel("Marks")

plt.show()
```

EXECUTED OUTPUT



RESULT

The Jupyter environment and required libraries were verified successfully through an interactive slider and an executed bar chart of student marks.